V_{ref}	V_{IN}	V_{OUT}	ΔV_{OUT}	I_{MAX}	ΔI_{OUT}	f_{sw}	η	D	datasheet
0.8 V	24 V	14.5 V	0.725V	3 A	1.5A	500 kHz	95%	0.604	link

Assuming: R10 = 10.2k

R9 = 174675 ≈ 174k 8.2.2.2 $\frac{Vout-Vre}{Vref} \times R10$

C14 = 100nF 8.2.2.3

K = 40% 8.2.2.5 $\frac{\Delta I_L}{I_{OUT\_MAX}}$

L1 = 9.57uH ≈ 10uH -> 15uH 8.2.2.5 $\frac{V_{IN} - V_{OUT}}{f_{SW} \times K \times I_{MAX}} \times \frac{V_{OUT}}{V_{IN}}$

But because in 8.2.2.6, Table 8-2 says for 12V output 12uH inductor is recommended I chose 15uH for my 14.5V output (webench has 10uH)

ΔI_L = 1.2 A $V_{IN\_MAX} = V_{IN}$ 8.2.2.5 $\frac{V_{IN\_MAX} - V_{OUT}}{L \times f_{SW}} \times \frac{V_{OUT}}{V_{IN\_MAX}}$

I_{L_PEAK} = 3.6A 8.2.2.5 $I_{OUT} + \frac{\Delta I_L}{2}$

I_{L_RMS} = 3.02A 8.2.2.5 $\sqrt{I_{MAX}{}^2 + \frac{\Delta I_L{}^2}{12}}$

$C_{OUT} \geq$ 5.93uF 8.2.2.6 $\frac{\Delta I_{OUT}}{f_{SW} \times \Delta V_{OUT} \times K} \times \left[(1-D) \times (1+K) + \frac{K^2}{12}(2-D)\right]$

C_{OUT} are C41 and C42 with 32uF in total

Input caps are from webench

Max current, assuming $R_{\theta JA}$ = 40 C/W (Junction- Board is 20 + 20 as copper plane to ambient) T_A = 25 and T_J = 100

I_{OUT_MAX} = 2.46 A 8.2.2.9 $\frac{(T_J - T_A)}{R_{\theta JA}} \times \frac{\eta}{1-\eta} \times \frac{1}{V_{OUT}}$

V_{IN}	V_{OUT}	I_{MAX}	f_{sw}	η	datasheet
14.5 V	7 V	2 A	650kHz	95%	link

Assuming: R13 = 10.2k

R12 = 83133 ≈ 84.5k 8.2.1.2.1 $\left(\frac{V_{OUT}}{0.765} - 1\right) \times R13$

L2 = 6.8 uH according to Table 2 in 8.2.1.2.2

I chose greater than 4.7uH to get lower current fluctuations

8.2.1.2.2

$I_{P\text{-}P}$ = 0,859 A $\frac{V_{OUT}(V_{IN(MAX)} - V_{OUT})}{V_{IN(MAX)} \times L_O \times f_{SW}}$

I_{PEAK} = 2.43 A $I_{MAX} + \frac{I_{P-P}}{2}$

$I_{lo(RMS)}$ = 2.02 A $\sqrt{{I_{MAX}}^2 + \frac{{I_{P-P}}^2}{12}}$

$I_{CO(RMS)}$ = 0.236 A $\frac{V_{OUT} \times (V_{IN} - V_{OUT})}{\sqrt{12} \times V_{IN} \times L_O \times f_{SW}}$

Rest is from datasheet

Input cap 8.2.1.2.3

Bootstrap cap 8.2.1.2.4